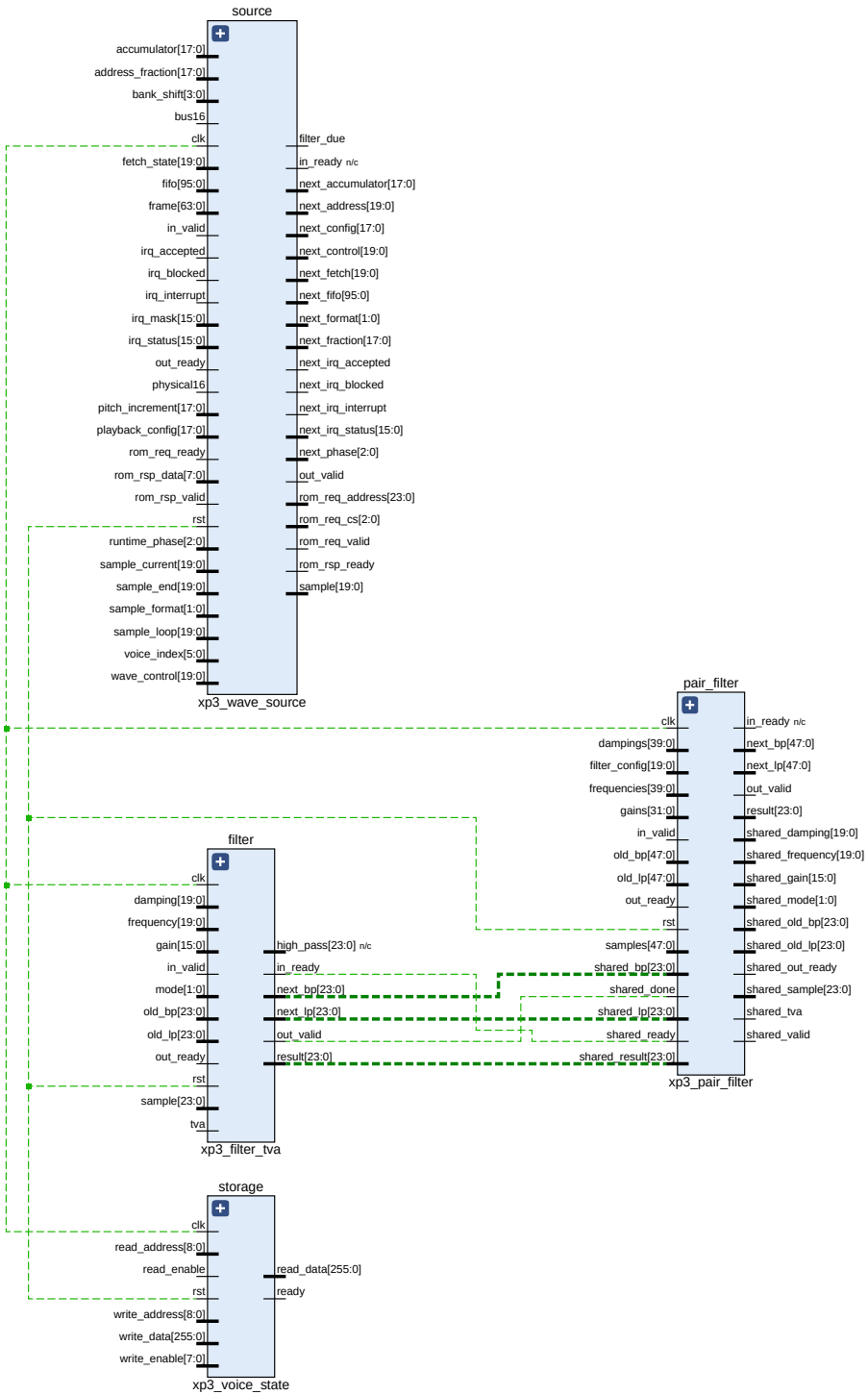
source
accumulator[17:0]
address_fraction[17:0]
bank_shift[3:0]
bus16
clk
fetch_state[19:0]
fifo[95:0]
frame[63:0]
in_valid
irq_accepted
irq_blocked
irq_interrupt
irq_mask[15:0]
irq_status[15:0]
out_ready
physical16
pitch_increment[17:0]
playback_config[17:0]
rom_req_ready
rom_rsp_data[7:0]
rom_rsp_valid
rst
runtime_phase[2:0]
sample_current[19:0]
sample_end[19:0]
sample_format[1:0]
sample_loop[19:0]
voice_index[5:0]
wave_control[19:0]
filter_due
in_ready n/c
next_accumulator[17:0]
next_address[19:0]
next_config[17:0]
next_control[19:0]
next_fetch[19:0]
next_fifo[95:0]
next_format[1:0]
next_fraction[17:0]
next_irq_accepted
next_irq_blocked
next_irq_interrupt
next_irq_status[15:0]
next_phase[2:0]
out_valid
rom_req_address[23:0]
rom_req_cs[2:0]
rom_req_valid
rom_rsp_ready
sample[19:0]
xp3_wave_source
pair_filter
clk
dampings[39:0]
filter_config[19:0]
frequencies[39:0]
gains[31:0]
in_valid
old_bp[47:0]
old_lp[47:0]
out_ready
rst
samples[47:0]
shared_bp[23:0]
shared_done
shared_lp[23:0]
shared_ready
shared_result[23:0]
in_ready n/c
next_bp[47:0]
next_lp[47:0]
out_valid
result[23:0]
shared_damping[19:0]
shared_frequency[19:0]
shared_gain[15:0]
shared_mode[1:0]
shared_old_bp[23:0]
shared_old_lp[23:0]
shared_out_ready
shared_sample[23:0]
shared_tva
shared_valid
xp3_pair_filter
filter
clk
damping[19:0]
frequency[19:0]
gain[15:0]
in_valid
mode[1:0]
old_bp[23:0]
old_lp[23:0]
out_ready
rst
sample[23:0]
tva
high_pass[23:0] n/c
in_ready
next_bp[23:0]
next_lp[23:0]
out_valid
result[23:0]
xp3_filter_tva
storage
clk
read_address[8:0]
read_enable
rst
write_address[8:0]
write_data[255:0]
write_enable[7:0]
read_data[255:0]
ready
xp3_voice_state